

Data Work in Finance: Finding, Downloading, Cleaning & Merging Data

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Roadmap

- ① **Why data is the binding constraint**
- ② **Where to find data** — vendors, free sources, HU (LEQR)
- ③ **Getting data out** — downloading, reshaping
- ④ **Cleaning & quality checks**
- ⑤ **Merging safely** — identifiers, keys, cardinality

Why data is the real constraint (not ideas)

- In finance research, **data availability** often determines what you can study.
- Financial data is frequently **expensive**, proprietary, or incomplete.
- Practical takeaway: start with a question you like, then **stress-test it** against what data you can obtain.

Rule of thumb

If you cannot name the dataset(s) and identifiers you will use, your research proposal is not ready.

Where data comes from

- Private data vendors (most common)
- Government agencies (often free, via websites, usually aggregate)
- Authors' websites (rare)
- Journals' replication packages (rare, becoming more common)

Finance reality check

You may need to **adjust the research question** to match feasible data.

When Your Key Variable Is Missing: CEO Experience & M&A

Goal: Study whether CEO experience affects M&A outcomes.

Problem: You *don't have CEO-level data* (no tenure, age, past deals, background).

What you can do instead:

- **Use firm-level proxies for “experience”**, maybe:
 - Past M&A activity (e.g., # of acquisitions in last 5 years)
 - Industry deal intensity
- **Shift the question slightly**, maybe:
 - “Do **firms with prior deal experience** have better announcement returns?”

Key lesson: If you can't observe the CEO, test the *mechanism* using a measurable proxy.

Databases at HU (LEQR)

Most licensed finance databases at HU are accessed via the **Humboldt Lab for Empirical and Quantitative Research (LEQR)**.

- You need an account (often arranged for a Bachelor thesis).
- Budget time: learning each interface is a project on its own.

Link

LEQR Data Access Portal

Main database families

Family	Typical content - NOT ALL AVAILABLE AT HU
Company fundamentals	Balance sheet / income statement / cash flows; annual/quarterly (Compustat, Capital IQ, WorldScope, Orbis/Amadeus, DAFNE)
Markets	Prices/returns/volume; bonds, CDS, rates, options (CRSP, Datastream, Bloomberg)
Ownership	Institutional and mutual fund holdings (13F-style; FactSet/Lipper/LSEG)
Deals & events	IPO/SEO, M&A, syndicated loans, VC/PE (LSEG Workspace/SDC, DealScan)
Other	Ratings, analyst forecasts (I/B/E/S), exec comp (Execucomp), ESG (LSEG Workspace)

Compustat: key identifier

- **GVKEY** = unique company identifier inside Compustat.
- Names (CONM) are convenient but **not reliable** (spelling, mergers, suffixes).
- Year can be **calendar** (CY) or **fiscal** (FY) — check carefully!

Common cross-database identifiers

CUSIP, SEDOL, ISIN, Ticker (often paired with a date).

LSEG Workspace: Deal Screener & Datastream

Deal Screener

M&A, equity/bond issues, syndicated loans, PE/VC deals. Stronger coverage for large firms and developed markets.

Datastream

Time series (prices, indices, rates).

Practical tips

Start Workspace **from the Excel add-in** to download directly to Excel. RICs work inside LSEG, but you often still need ISIN / Ticker / CUSIP to merge with other databases.

Macroeconomic data (free)

- World Bank: World Development Indicators (WDI)
- IMF: data portal
- National statistical offices / central banks

Use:

Combine firm outcomes with country-level variables (GDP growth, inflation, CO2 emissions, etc.).

EDGAR (SEC filings)

- U.S. listed firms: filings (10-K, 10-Q, 8-K, proxy statements, etc.).
- Great for **text analysis**, accounting footnotes, event dates, insider trading, ownership, etc.

Link
SEC EDGAR

Data management

A simple project folder structure

```
project/  
  data_raw/           # vendor exports, never edited  
  data_intermediate/  # cleaned but not final  
  data_final/         # analysis-ready datasets  
  code/               # scripts (download/clean/merge/analysis)  
  output/             # tables, figures, logs  
  docs/               # data dictionary, notes
```

- **Unit of observation:** firm-year, stock-day, loan-quarter, ...
- **Save query parameters** so the download is reproducible.
- **Clean in code**, not by hand-editing spreadsheets.
- **Don't set missing to 0** unless 0 truly means zero (and justify it).

Cleaning & quality checks

1. Standardize

- Types: dates, numbers, strings
- Units: currency, bps, frequency
- Outliers: investigate, then winsorize
- Dates: fiscal vs calendar

2. Check

- Sanity: min/max, unique IDs, ranges
- Duplicates: do keys identify rows?
- Economic sense: prices > 0 , returns
- Log every drop / keep

Worked examples & details in the **Appendix**.

Cleaning: spot the problems

	gvkey	conm	fyear	datadate	industry	sales	ret
1	1001	Acme Corp	2022	2022-12-31	Tech	1,200	0.05
2	1001	Acme Corp	2022	2022-12-31	Tech	1,200	0.05
3	1002	Globex	2022	12/31/2022	tech	\$3,400	12%
4	1003	Initech	2022	2022-12-31	TECH	980	NA
5	1004	Umbrella	2022	2022-12-31	Retail	2,500	-0.08
6	1004	Umbrella	2023	2023-12-31	retail	2,750	0.10

Orange box: Duplicate row (1 = 2): firm-year not unique

Yellow box: Inconsistent date format (US vs ISO)

Blue box: Inconsistent labels: Tech/tech /TECH, Retail/retail

Green box: Numbers stored as text (, and \$)

Red box: Missing as text ("NA") & mixed units ("12%")

Cleaning: the same data, fixed

After cleaning: duplicate dropped (6 -> 5 rows), firm-year now unique

	gvkey	conm	fyear	datadate	industry	sales	ret
1	1001	Acme Corp	2022	2022-12-31	Tech	1200	0.05
2	1002	Globex	2022	2022-12-31	Tech	3400	0.12
3	1003	Initech	2022	2022-12-31	Tech	980	. (missing)
4	1004	Umbrella	2022	2022-12-31	Retail	2500	-0.08
5	1004	Umbrella	2023	2023-12-31	Retail	2750	0.10

■ Fixed: dates parsed, labels standardized, numbers numeric, 12% -> 0.12, NA -> missing (.)

Merging: how to avoid silent disasters

1) Define your merge key(s)

Examples: GVKEY + FY, ISIN + date, CUSIP6 + month.

2) Check merge cardinality

Cardinality = how many rows on each side share a key value: one-to-one (ideal)? one-to-many (e.g. firm + multiple bonds; date + intraday trades)? many-to-many (very dangerous! — fix IDs, duplicates, or aggregate first).

3) Audit the result

Report match rates; inspect the *non-matches*; save a merge log.

Merging: the two files

Merge File A + File B on **gvkey** (each firm-year gets its firm info)

File A: firm-years (the data) - key: **gvkey**

	gvkey	conm	fyear
1	1001	Acme Corp	2022
2	1002	Globex	2022
3	1003	Initech	2022
4	1004	Umbrella	2022
5	1004	Umbrella	2023

File B: firm reference table - key: **gvkey**

	gvkey	country	hq_city
1	1001	USA	New York
2	1002	USA	Chicago
3	1004	UK	London
4	1005	Germany	Berlin

1004 appears twice in A -> one-to-many

1003 is in A but NOT in reference table B

1005 is in B but NOT in the data A

Merging: the result

Result: merge firm-years + reference table (key = gvkey, m:1) - match rate 4 of 5

	gvkey	conm	fyear	country	_merge
1	1001	Acme Corp	2022	USA	3 matched
2	1002	Globex	2022	USA	3 matched
3	1003	Initech	2022	(missing)	1 data only
4	1004	Umbrella	2022	UK	3 matched
5	1004	Umbrella	2023	UK	3 matched
6	1005	(no firm-year)	--	Germany	2 reference only

- One-to-many: 1004 has two firm-years
- Non-match: 1003 in data, not in reference table (_merge=1)
- Unused: 1005 in reference table, no firm-year (_merge=2)

Identifiers in practice: what usually works

- Best: **stable vendor identifiers** (GVKEY, PERMNO, ISIN), often combined with date.
- Good: **ticker + date** (tickers change!) or **CUSIP + date** (mind CUSIP digits!).
- Last resort: **names** $\Rightarrow$ fuzzy matching.

Fuzzy matching

Normalize names (case, punctuation, suffixes, accents), match within country/industry, and always validate a random sample by hand. (Details & examples in the **Appendix**.)

Quick exercise: which merge key?

You want **monthly returns alongside each firm's annual credit rating**:

- File A: monthly stock returns — one row per **firm-month**
- File B: credit ratings — one row per **firm-year**

What is the merge key? What is the cardinality?

Key: `firm + year` (map each month to its year). Cardinality: **many-to-one** — 12 firm-months to one firm-year. Merging on `firm` alone would be many-to-many: a silent duplication bug.

Takeaways

- Data work is **a considerable amount if not most of the work** in empirical finance.
- Track identifiers and frequencies.
- Keep raw data raw; make cleaning reproducible.
- Merges need identifiers and checks.

Appendix: Cleaning examples

Standardize types

- Dates: 2024-03-01 vs 03/01/2024 (US) vs 01/03/2024 (EU)
- Numbers: "1,200" or "\$1.2m" → 1200 or 1200000
- Strings: "Tech", "technology", "TECH " → one value (casing/spaces)

Standardize units

- Returns: 5% = 0.05 (not 5)
- Rates: 250 bps = 2.5% = 0.025
- Scale: revenue in **thousands** vs assets in **millions** ⇒ wrong ratios

Outliers & dates

- A 300% return could be a *true* event — investigate before winsorizing.
- FY2023 financials may be released in 2024 (fiscal vs calendar year).

Appendix: Fuzzy matching tips

- **Normalize names:** "Apple Inc." → "apple"; remove punctuation/suffixes: "Ltd", "PLC", "S.A.", "Corp".
- **Fix spacing/accents:** "Banco Santander" → "banco santander"; "Energía" → "energia".
- **Match within groups:** compare only firms in the same **country** and/or **industry** (e.g., US banks vs US banks) to avoid false matches.
- **Validate manually:** check 20–50 random matches and flag suspicious ones (e.g., "Meta Platforms" vs "MetLife").